FedSemGNN System Architecture

Legend
Metrics
Hardware
Execution
Federation
PPO
GCN
Semantic

Ingestion
Clients / Requests
requests
JSON/Protobuf
Semantic Processing (extract vector)
semantic vector
ℝ^d

Learning
GCN Encoder (embeddings)
embeddings
node × hid
PPO Agents (policy/value)
global update
θ
local params
θ_i
decision
argmax π(a|s)
Federated Sync (FedAvg + DP)

Execution
Provisioning (place service)
emit
Metrics Logger (latency, fidelity, power, bytes)
deploy
Edge Servers / Network (resources, power)